UNIT - IV

Design of Clutches, Brakes and Springs

1. Friction Clutches

A clutch connects / disconnects a driven shaft from a driving shaft (engine - gearbox) while running, by friction between surfaces pressed together by an axial force W. It must engage smoothly (slip, then lock), transmit the full torque without slip, and dissipate heat.

- Types : positive (jaw, dog, spline - no slip, engage at rest) ; friction (disc / plate - single & multi plate, cone, centrifugal) ; hydraulic / fluid (fluid coupling) ; electromagnetic.
- Friction material : lining woven asbestos-free / resin-bonded organic (μ = 0.3-0.4), sintered metal (0.15-0.4), cork (0.3-0.35, wet), leather (0.3-0.5) on C.I. or steel ; p_{max} = 0.1-0.5 N/mm² (dry), up to 1 N/mm² (wet).
- Requirements : high & stable μ, wear and heat resistance, good thermal conductivity, cheap, no fade, quick cooling, small inertia, balanced.

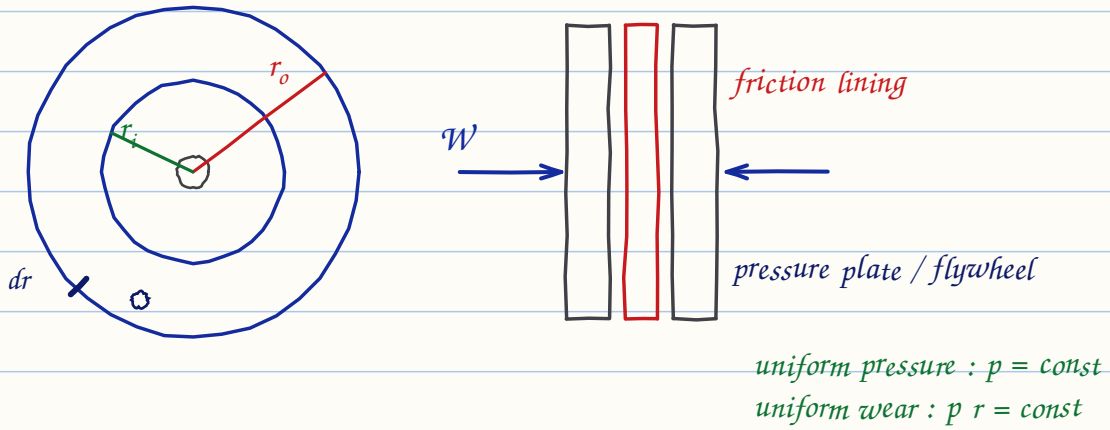


Fig 1: Single-plate (disc) clutch and pressure laws

Torque transmitting capacity - disc clutch

Annular lining : outer radius r_o, inner r_i, axial force W, coefficient μ. Take an element of radius r, width dr. Normal force $dW = p\ 2\pi\ r\ dr$; friction torque $dT = \mu\ p\ 2\pi\ r^2\ dr$.

(a) Uniform pressure theory (new clutch) : p = const

$$W = \pi\, p\, (r_o^2 - r_i^2)$$
$$T = (2/3)\,\pi\,\mu\, p\, (r_o^3 - r_i^3) = \mu\, W\, R_m$$
$$R_m = (2/3)\,(r_o^3 - r_i^3)/(r_o^2 - r_i^2)$$

(b) Uniform wear theory (worn-in clutch) : $p\,r = C$ = const

Wear $\propto p \times v \propto p\,r$ (since $v = \omega\, r$), so for equal wear $p\,r = C$; maximum pressure is at the inner radius : $C = p_{max}\, r_i$.

$$W = 2\pi\, C\, (r_o - r_i) = 2\pi\, p_{max}\, r_i\, (r_o - r_i)$$
$$T = \pi\,\mu\, C\, (r_o^2 - r_i^2) = \mu\, W\, R_m\,,\quad R_m = (r_o + r_i)/2$$

- For the same W : T(uniform pressure) > T(uniform wear) (R_m larger) , so a clutch is designed with the uniform wear value (lower, conservative) as the lining wears after the initial run-in.
- Multi-plate clutch : n_1 discs on driving shaft , n_2 on driven → number of pairs of contact surfaces $n = n_1 + n_2 - 1$: $T = n\,\mu\, W\, R_m$. Large torque in small diameter (motorcycles, wet clutch in oil, automatic transmission).
- For maximum torque (given r_o, p_{max}, uniform wear) : $dT/dr_i = 0 \rightarrow r_i = r_o/\sqrt{3} = 0.577\, r_o$. Usually $r_i/r_o = 0.6$-0.7 (practical).
- Power : $P = T\,\omega = 2\pi\, N\, T / 60$; design torque $T_d = K_s \times T_{engine}$ ($K_s = 1.25$-2.0). Engaging spring force = W (diaphragm spring in cars).

Example 1 : Single-plate clutch with both sides effective. $r_o = 125$ mm, $r_i = 75$ mm, $\mu = 0.3$, $p_{max} = 0.1$ N/mm², 1500 rpm. Uniform wear : $W = 2\pi \times 0.1 \times 75 \times 50 = 2356$ N ; $R_m = 100$ mm ; $T = n\,\mu\, W\, R_m = 2 \times 0.3 \times 2356 \times 100 = 141 \times 10^3$ N·mm = 141 N·m → $P = 141 \times (2\pi \times 1500/60) = 22.2$ kW.

(Uniform pressure, same p : $W = \pi \times 0.1 \times (125^2 - 75^2) = 3142$ N ; $R_m = 102$ mm ; $T = 2 \times 0.3 \times 3142 \times 102 = 192$ N·m - 36% more - not safe to use.)

Heat generated & slip time (clutch engagement)

$$\text{Slip energy}\quad E = \tfrac{1}{2}\, I_e\, \omega_1^2\,,\quad I_e = I_1\, I_2/(I_1 + I_2)$$
$$\text{Slip time}\quad t = I_2\, \omega_1 / (T - T_{load})$$
$$\text{Heat flux}\quad q = E / (A\, t)\quad (W/mm^2)$$

- Smooth engagement needs t = 0.5-1.5 s for cars. Too short → jerk (shock load on shaft), too long → overheating of lining (fade, glazing). Limit $q \approx 0.4\text{-}1.0\ W/mm^2$ for dry linings. Temperature rise $\Delta T = E / (m\ c)$.
- Positive (jaw / claw) clutch : square or spiral jaws ; torque by shear / bearing on the jaw face : $T = n\ (A_{jaw}\ \tau)\ R_m$; engaged only at rest or very low relative speed (no slip - shock).

2. Cone Clutch

Friction surfaces are conical (semi-cone angle $\alpha \approx 10°$ to $12.5°$; $\alpha > 8.5°$ so that the cone does not jam / self-lock : $\tan \alpha > \mu$). Wedge action gives large normal force with small axial W :

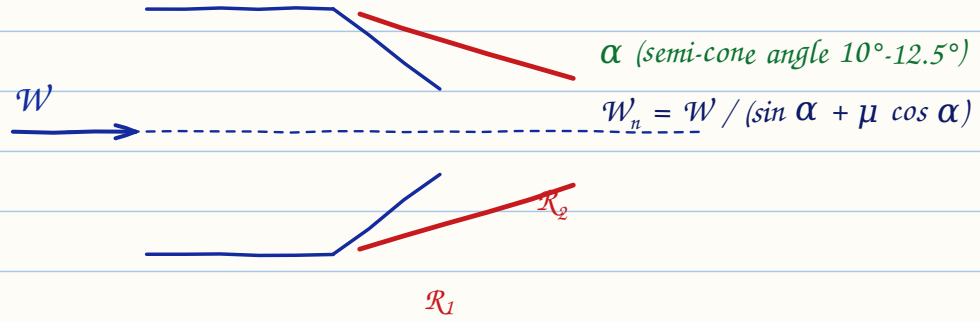


Fig 2: Cone clutch

Engagement : $W = W_n\ (\sin \alpha + \mu \cos \alpha)$ (W_n = normal force on cone)

Running (uniform wear) : $T = \mu\ W_n\ R_m = \mu\ W\ R_m / \sin \alpha$

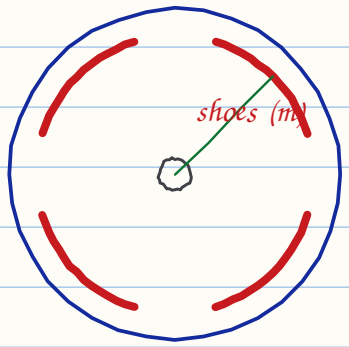
Uniform pressure : $R_m = (2/3)(R_1^3 - R_2^3)/(R_1^2 - R_2^2)$; $W = \pi\ p\ (R_1^2 - R_2^2)$

Uniform wear : $R_m = (R_1 + R_2)/2$

- Face width b along the cone ; $R_1 - R_2 = b \sin \alpha$. Pressure $p = W_n / (2\pi\ R_m\ b)$. Torque capacity ≈ 5-8 times more than a plate clutch of equal size for the same W ($\alpha = 12.5°$) - but needs accurate alignment, sticks if α too small, disengagement needs force W_{dis}.

3. Centrifugal Clutch

Shoes with friction lining are pivoted on a driving spider and are held inward by springs. When the speed rises above the engagement speed ω_1 the centrifugal force overcomes the spring and the shoes press on the drum - automatic clutch (mopeds, chain-saws, go-karts, starting of motors).



at speed ω :

centrifugal $P_c = m\,\omega^2\,r$

spring $P_s = m\,\omega_1^2\,r$ (ω_1 = engagement)

normal $N = P_c - P_s$, $F = \mu N$

$T = n\,\mu\,(P_c - P_s)\,R$

Fig 3: Centrifugal clutch

$P_c = m\,\omega^2\,r$, $P_s = m\,\omega_1^2\,r$ (r = c.g. radius of shoe when engaged)

Normal on drum : $N = P_c - P_s = m\,r\,(\omega^2 - \omega_1^2)$

$T = n\,\mu\,N\,R = n\,\mu\,m\,r\,(\omega^2 - \omega_1^2)\,R$

- Usually the shoes begin to engage at $\omega_1 = 0.75\,\omega$ (¾ running speed) → $P_s = (9/16)\,P_c$; $N = (7/16)\,m\,r\,\omega^2$. Pressure on lining : $p = N / (b\,l)$, $l = R\theta$ (shoe arc) , $p \le 0.1$-0.2 N/mm².

Example 2 : 15 kW, 900 rpm, 4 shoes, drum R = 150 mm, shoe c.g. r = 120 mm, μ = 0.25, $\omega_1 = 0.75\,\omega$. $T = 9550 \times 15/900 = 159$ N·m. $N = T/(n\,\mu\,R) = 159/(4 \times 0.25 \times 0.15) = 1061$ N. With $\omega = 94.25$ rad/s : $N = (7/16)\,m\,r\,\omega^2 \rightarrow m = 1061/(0.4375 \times 0.12 \times 8883) = 2.27$ kg per shoe. If $\theta = 60°$, b = 40 mm : l = 157 mm → $p = 1061/(157 \times 40) = 0.17$ N/mm² (OK).

4. Brakes - General

A brake absorbs the kinetic / potential energy of a moving system by friction and dissipates it as heat, to slow or stop (or hold) the system.

Energy to be absorbed : $E = \tfrac{1}{2}\,I\,(\omega_1^2 - \omega_2^2) + \tfrac{1}{2}\,m\,(v_1^2 - v_2^2) + m\,g\,h$

Braking torque : $T_b = \mu\,N\,r_d$; heat generated $Q = T_b\,\theta$ (θ = angle turned in braking)

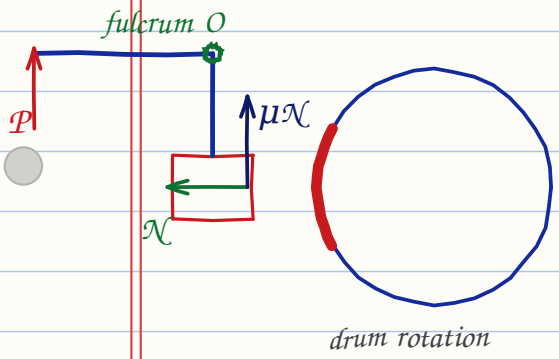
Temperature rise : $\Delta T = E / (m_{drum}\,c)$ ($c \approx$ 460-500 J/kg K steel / C.I.)

Brake	Description	Use
Block / shoe (external)	single / double rigid or pivoted shoe pressed on drum	railway, cranes, lifts
Band	flexible steel band with lining wrapped on drum	hoists, excavators, chain saw
Internal expanding	two shoes inside drum, expanded by cam / hydraulic cylinder	car, motorcycle rear
Disc (caliper)	pads squeezed on a rotating disc	cars, aircraft, bikes
Cone / Axial	like cone clutch	winches

- Brake materials : drum C.I. FG 260 ; shoe lining asbestos-free Ferodo / woven cotton / sintered, μ = 0.3-0.45, p_{max} = 0.3-1 N/mm² ; band : mild steel, lining leather / fabric. Brake must also be cooled (fins, ventilation).

5. Block (Shoe) Brake

Single block brake : block pressed on drum by lever ; normal reaction N, friction $F = \mu N$ tangential, braking torque $T_b = \mu N r$.



$P l = N a \mp \mu N c$

friction moment aids P → self-energizing

if $a \le \mu c$ → $P \le 0$: self-locking

(c = offset of fulcrum from the friction line)

Fig 4: Single block brake with offset fulcrum

$P l = N a - \mu N c$ (friction moment assists P) → self-energizing

$P l = N a + \mu N c$ (rotation reversed → friction opposes)

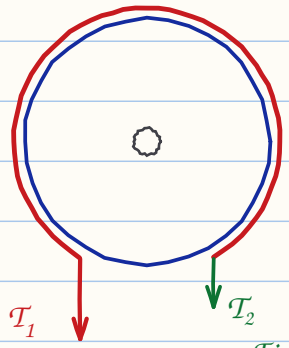
$T_b = \mu N r$, $N = P l / (a \mp \mu c)$

- Self-energizing brake : friction force itself helps to apply the brake → smaller P for the same torque (mechanical advantage ↑). Self-locking : when $a \le \mu c$, P becomes zero or negative - the brake grabs & cannot be released (used deliberately in some ratchets, but dangerous in vehicles).
- Pivoted block ($\theta > 60°$) : equivalent coefficient $\mu' = 4 \mu \sin(\theta/2) / (\theta + \sin\theta)$ (θ in rad) ; pressure uniform, no tilting moment. Double-block brake balances the shaft bearing load (no side thrust).

- Block pressure $p = N / (w \times 2\,r \sin(\theta/2)) \le 0.5\text{-}1\ N/mm^2$; heat per unit area per second $\mu\ p\ v \le 0.3\text{-}0.5\ N{\cdot}m/mm^2{\cdot}s$ (limit of heating).

6. Band Brake

The flexible band on drum : tension T_1 (tight) and T_2 (slack), with the Eytelwein relation for belt friction :



$T_1/T_2 = e^{\mu\theta}$

$T_b = (T_1 - T_2)\ r$

$p_{max} = 2\ T_1 / (w\ D)$

simple : $P\ l = T_2\ b$ (slack end on lever)

differential : $P\ l = T_2\ b_2 - T_1\ b_1$

Fig 5: Band brake - simple and differential

- Drum radius effective $r = r_d + t/2$ (t = band thickness). Wrap $\theta = 180°\text{-}270°$, $\mu = 0.2\text{-}0.4$ (leather / fabric on steel).
- Simple band : lever force $P = T_2\ b / l$ when slack end is on the lever (large wrap helps) ; if tight end T_1 on lever : $P = T_1\ b / l$ (twice or more).
- Differential band : tight and slack ends on opposite sides of fulcrum ; $P\ l = T_2\ b_2 - T_1\ b_1 = T_2\ (b_2 - b_1\ e^{\mu\theta}) \rightarrow$ self-locking if $b_2 \le b_1\ e^{\mu\theta}$; low P but unsafe for reversal.
- Band design : width w and thickness t from tension ; $\sigma_t = T_1 / (w\ t) \le 50\text{-}80$ MPa (with rivet-hole weakening : $T_1/((w - n\ d_r)\ t)$) ; max pressure at tight end $p_{max} = 2\ T_1/(w\ D) \le 0.3\text{-}0.5\ N/mm^2$.

Example 3 : Drum $D = 400$ mm, $T_b = 400\ N{\cdot}m$, wrap 240° (4.19 rad), $\mu = 0.25$. $e^{\mu\theta} = e^{1.047} = 2.85$. $T_1 - T_2 = 400/0.2 = 2000\ N \rightarrow T_2 = 2000/1.85 = 1082\ N$, $T_1 = 3082\ N$. Simple band, $b = 60$ mm, $l = 600$ mm : $P = T_2\ b / l = 1082 \times 60/600 =$ 108 N. Band $w = 50$, $t = 3$: $\sigma = 3082/(50 \times 3) = 20.5$ MPa (OK) ; $p_{max} = 2 \times 3082/(50 \times 400) = 0.31\ N/mm^2$ (OK).

7. Internal Expanding Shoe & Disc Brake

Internal expanding shoe brake (drum brake)

- Shoe pivoted at one end, expanded against rotating drum by cam / wheel cylinder. *Leading shoe* : drum drags shoe into the drum - friction moment M_f assists (self-energizing) ; *trailing shoe* : friction opposes.

$$\text{Actuating force : } F = (M_N - M_f)/c \quad \text{(leading)} \;;\; F = (M_N + M_f)/c \quad \text{(trailing)}$$
$$\text{Torque per shoe : } T = \mu\, p_{max}\, b\, r^2 (\cos\theta_1 - \cos\theta_2)/\sin\theta_a$$

- M_N = moment of normal forces about pivot, M_f = moment of friction forces ; self-locking when $M_f \geq M_N$. Pressure varies as $\sin\theta$ (zero at the pivot). Dual-leading arrangement (2 leading shoes) increases torque ; hydraulic cylinder gives equal force on both shoes.

Disc brake

Annular pads (sector of angle θ) clamped on a rotating disc by caliper pistons ; both sides of the disc work. Uniform wear used for design :

$$F = p_{max}\, r_i\, \theta\, (r_o - r_i) \quad \text{(per pad)}$$
$$T_b = n\,\mu\,F\,R_m = n\,\mu\,p_{max}\,\theta\,r_i\,(r_o^2 - r_i^2)/2\;,\quad R_m = (r_o + r_i)/2$$

- *Advantages* over drum : no fade (open to air, self-cleaning, ventilated discs), linear torque vs pedal force, uniform wear, easy pad change, stable in wet, balanced axial thrust. *Disadvantages* : needs high hydraulic pressure (servo), pad area small → pad wear faster, costly.

Example 4 : A 1000 kg vehicle at 72 km/h (20 m/s) stops. $E = ½ \times 1000 \times 20^2 = 200$ kJ. If the four brake drums have 20 kg mass each (80 kg total, $c = 480$ J/kg K) → $\Delta T = 200000/(80 \times 480) = 5.2$ °C for a single stop ; 25 repeated stops → fade. Disc pad ($n = 2$, $\mu = 0.35$, $p_{max} = 1$ N/mm², $\theta = 60° = 1.05$ rad, $r_o = 120$, $r_i = 70$) : $F = 1 \times 70 \times 1.05 \times 50 = 3675$ N ; $T = 2 \times 0.35 \times 3675 \times 95 = 244$ N·m per disc.

8. Helical Springs - Compression & Tension

A *spring* stores energy and releases it (shock absorption, force application, energy storage, measurement). Helical spring of round wire is loaded axially - the wire experiences *torsion + direct shear*.

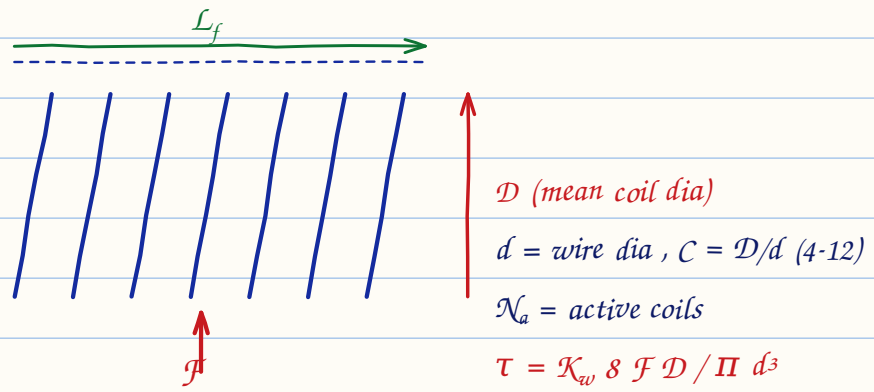


Fig 6: Helical compression spring terminology

- Spring index $C = D / d$ (4-12 preferred ; < 4 hard to coil, > 12 tangles / buckles). Active coils N_a ; total N_t ; pitch p ; free length L_f ; solid length L_s ; stiffness (rate) $k = F/\delta$.

End type	N_t	Solid length L_s
Plain ends	N_a	$(N_t + 1)\ d$
Plain & ground	$N_a + 1$	$N_t\ d$
Squared (closed)	$N_a + 2$	$(N_t + 1)\ d$
Squared & ground	$N_a + 2$	$N_t\ d$

$$\tau = K_w \cdot 8\ F\ D\ /\ \pi\ d^3 = K_w\ 8\ F\ C\ /\ \pi\ d^2$$
$$K_w = (4C - 1)/(4C - 4) + 0.615/C \quad \text{(Wahl factor)}$$
$$\delta = 8\ F\ D^3\ N_a\ /\ G\ d^4 = 8\ F\ C^3\ N_a\ /\ G\ d$$
$$k = F/\delta = G\ d^4\ /\ 8\ D^3\ N_a \quad , \quad U = \tfrac{1}{2}\ F\ \delta$$

- Wahl factor K_w accounts for direct shear $(1 + 0.5/C)$ and curvature of the wire (higher stress on the inside of the coil). K_w = 1.1-1.6 for C = 12-4. For static load on ductile wire $K_s = 1 + 0.5/C$ may be used ; for fatigue always K_w.
- Free length : $L_f = L_s + \delta_{max}$ + clearance (≈ 15% δ_{max} or 1-2 mm per coil). Buckling check : L_f/D < 2.6 (ends fixed, no guide) ; otherwise use a guide rod / tube. Surge : natural frequency $f_n = \tfrac{1}{2}\sqrt{(k / m_s)} \geq 13 \times$ operating frequency.
- Materials : hard-drawn (A227), oil-tempered (A229), music wire (A228 - best, $d \leq 6$ mm), chrome-vanadium (A231, high temp / fatigue), 55Si2Mn90 / 60Si2Mn (IS) for heavy springs, phosphor bronze, stainless. $\tau_{allow} = 0.45\text{-}0.5\ S_{ut}$ (static) ; G = 80-84 GPa (steel).

$$d = \sqrt{(8\,K_w\,F\,C/\pi\,\tau_{allow})} \quad ; \quad N_a = \delta\,G\,d/(8\,F\,C^3)$$ *(design steps)*

Example 5 : Spring for $F = 1000$ N and $\delta = 25$ mm, $C = 6$, $\tau_{allow} = 420$ MPa, $G = 84$ GPa. $K_w = 23/20 + 0.615/6 = 1.253$. $d = \sqrt{(8 \times 1.253 \times 1000 \times 6/(\pi \times 420))} = 6.75 \rightarrow d = 7$ mm. $D = 6 \times 7 = 42$ mm. $N_a = 25 \times 84000 \times 7/(8 \times 1000 \times 216) = 8.5 \rightarrow 9$; squared & ground $N_t = 11$, $L_s = 11 \times 7 = 77$ mm , $L_f = 77 + 25 + 4 = 106$ mm , $k = 40$ N/mm.

Springs in combination

Parallel (concentric) : $k = k_1 + k_2$, $F = F_1 + F_2$, same δ

Series : $1/k = 1/k_1 + 1/k_2$, same F , $\delta = \delta_1 + \delta_2$

- *Concentric springs* : same free length & deflection, opposite hand (no jamming), $D_{outer,inner}$ chosen with radial clearance ; $F_1/F_2 = k_1/k_2$; for equal τ, C equal $\rightarrow$ $d_1/d_2 = D_1/D_2$.

Fluctuating load (fatigue of springs)

$$F_m = (F_{max} + F_{min})/2 \;,\; F_a = (F_{max} - F_{min})/2$$

$$\tau_a = K_w\,8\,F_a\,D/\pi\,d^3 \;,\; \tau_m = K_s\,8\,F_m\,D/\pi\,d^3$$

$$1/n = \tau_a/S_{se} + \tau_m/S_{sy} \quad \text{(Soderberg - shear)}$$

$S_{sy} = 0.577\,S_y \approx 0.45\,S_{ut}$; S_{se} (torsional endurance, 10^7 cycles) ≈ 0.2-$0.3\,S_{ut}$ (unpeened) - shot peening improves by ~ 20%. Hooks of tension springs are weak points : stress concentration ($K = (4C_1^2 - C_1 - 1)/4C_1(C_1 - 1)$).

Tension (extension) spring - special points

- Wound with *initial tension* F_i (coils pressed together) - spring starts to extend only when $F > F_i$; $F = F_i + k\delta$. Body length $L_b = (N_a + 1)\,d$; free length (with hooks) $L_f = L_b + 2\,(D - d)$ (full loop).

9. Torsion Spring

Helical wire acted on by a *moment* $M = F\,r$ about the coil axis (door hinge, clothes peg, garage door) - the wire is in *bending*, not torsion. Moment tends to close (wind up) the coils.

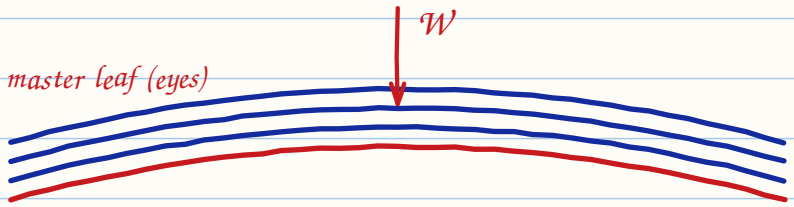
$$\sigma = K_i \cdot 32\,M / \pi\,d^3 \quad , \quad K_i = (4C^2 - C - 1)/(4C\,(C - 1)) \ \text{(inner fibre)}$$
$$\theta = 64\,M\,D\,N_a / E\,d^4 \ \text{(rad)} \quad = \quad 10.2\,M\,D\,N_a / E\,d^4 \ \text{(revolutions)}$$

- Direction of winding : load should tend to wind the spring up (increase the number of turns) → residual stress opposes working stress. Mean dia reduces as $D' = D\,N_a/(N_a + \theta_{rev})$ - leave clearance over the guide rod (≥ 10% of d).
- Stiffness (per radian) = $M/\theta = E\,d^4 / 64\,D\,N_a$; ends : straight, hinged, short or long legs ; $N_a = N_{body} + (L_1 + L_2)/(3\pi D)$ for the legs. Use $\sigma_{allow} = 0.7\,S_{ut}$ (static, cold wound, stress-relieved).

Example 6 : torsion spring $M = 3000$ N·mm, wire $d = 3$ mm, $C = 8$ ($D = 24$ mm), $E = 207$ GPa. $K_i = (256 - 8 - 1)/(4\times8\times7) = 1.10$; $\sigma = 1.10 \times 32 \times 3000/(\pi \times 27) = 1244$ MPa (within music wire's $0.7\,S_{ut}$). θ per turn $= 10.2 \times 3000 \times 24\,N_a/(207000 \times 81) \rightarrow$ for $N_a = 6$: 0.263 rev = 95°.

10. Leaf Springs

Semi-elliptic leaf spring (automobile & railway suspension) : several flat strips (leaves) of decreasing length clamped together ; acts as a beam of uniform strength. Interleaf friction gives damping.



semi-elliptic leaf spring , span 2L
each half = cantilever of length L , load P = W/2

Fig 7: Semi-elliptic multi-leaf spring

- Equivalent : a triangular plate of width n b and thickness t, cantilever length L = half span, end load P = W/2 → uniform bending stress along the length (graduated leaves). Full-length (extra) leaves n_f (master leaf with eyes) take shear, side loads ; they are over-stressed so a nip (initial gap C) is given for pre-stressing - gradually equalising stress.

Graduated leaves only : $\sigma = 6\,P\,L / n\,b\,t^2$, $\delta = 6\,P\,L^3 / E\,n\,b\,t^3$

With n_f full-length + n_g graduated leaves :

$\sigma_g = 12\,P\,L / b\,t^2\,(2n_g + 3n_f)$ $\quad \sigma_f = 18\,P\,L / b\,t^2\,(2n_g + 3n_f)$

$\delta = 12\,P\,L^3 / E\,b\,t^3\,(2n_g + 3n_f)$

- Load share : $P_g = 2\,n_g\,P / (2n_g + 3n_f)$; $P_f = 3\,n_f\,P / (2n_g + 3n_f)$; $\sigma_f = 1.5\,\sigma_g$ (hence nip). Nip $C \approx 2\,P_f\,L^3$ … chosen so that after bolting $\sigma_f = \sigma_g$. Spring rate $k = W/\delta_{total}$.
- Materials : silicon-manganese / chrome-vanadium steel (55Si2Mn90 , 50Cr1V23) ; oil hardened & tempered, shot peened ; σ_{allow} = 400-500 MPa ; leaves lubricated graphite. Fatigue : K_f at eye, centre bolt, clip - use rolled / shot peened.

Example 7 : load W = 12 kN, span 1000 mm → L = 500 , P = 6 kN. n_f = 2, n_g = 6 , b = 50 , σ_{allow} = 420 MPa. $\sigma_f = 18 \times 6000 \times 500/(50\,t^2 \times 18) = 60000/t^2 \le 420 \rightarrow t$ = 12 mm. σ_g = 278 MPa , σ_f = 417 (OK). $\delta = 12 \times 6000 \times 500^3/(207000 \times 50 \times 12^3 \times 18)$ = 28 mm (≈ 29 mm incl. camber).

Typical exam questions - Unit IV

- (1) Derive torque capacity of a disc clutch by uniform pressure and uniform wear theory ; which is used for design and why ?
- (2) Design a multi-plate clutch for given power, speed, p_{max}, μ ; find number of plates.
- (3) Explain self-energizing and self-locking of brakes ; condition for a block brake with offset fulcrum.
- (4) Derive $T_1/T_2 = e^{\mu\theta}$ and design a band brake ; differential band brake self-locking condition.
- (5) Design a helical compression spring for load F and deflection δ ; define Wahl factor, spring index.
- (6) Torsion spring stress and deflection ; design of semi-elliptic leaf spring with full-length leaves and nip.

Spring	Stress	Deflection
Helical compression / tension	$\tau = K_w\ 8FD/\pi d^3$	$\delta = 8FD^3N_a/Gd^4$
Torsion	$\sigma = K_i\ 32M/\pi d^3$	$\theta = 64\ M\ D\ N_a/E\ d^4$
Leaf (graduated)	$\sigma = 6PL/nbt^2$	$\delta = 6PL^3/Enbt^3$
Clutch (uniform wear)	$T = \mu\ W\ (r_o+r_i)/2$	$n = n_1+n_2-1$
Band brake	$T_1/T_2 = e^{\mu\theta}$	$T_b = (T_1-T_2)\ r$